University of Sheffield

Ancillary Material:
None

SCHOOL OF COMPUTER SCIENCE
Autumn Semester 2024–2025

Module Code and Title:
COM2004: Data Driven Computing

Exam Duration:
2 hours

Exam Instructions:

Answer THREE questions.

All questions carry equal weight. Figures in square brackets indicate the percentage of available marks allocated to each part of a question.

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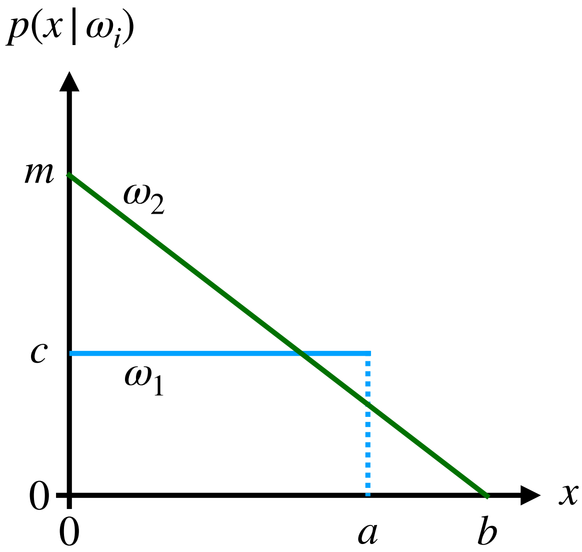
1. **Bayesian Decision Theory**

Consider a classification task based on the likelihood distributions sketched below



These likelihoods are defined by the following functions

$$p(x|\omega_1) = \begin{cases} 0 & x < 0 \\ c & 0 \leq x \leq a \\ 0 & a < x \end{cases}, \qquad p(x|\omega_2) = \begin{cases} 0 & x < 0 \\ m\left(1 - \frac{x}{b}\right) & 0 \leq x \leq b \\ 0 & b < x \end{cases}$$

where a, b, c and m are constants.

a) By considering that the likelihood for each class must be normalised, show that $c = \frac{1}{a}$ and $m = \frac{2}{b}$. **[5%]**

b) Samples for both classes were collected with the following values for x:

$$\omega_1 : 0.1,\ 0.7,\ 2$$
$$\omega_2 : 0.2,\ 1.8,\ 3$$

The maximum likelihood estimates for a and b from this data are $a = 2$ and $b = 3$, give an intuitive explanation for why this is the case. **[10%]**

c) Assume that the classes are equally likely, i.e that they have equal priors. Using the values of $a = 2$ and $b = 3$, specify the ranges of x where each class ω_i, where $i \in \{1, 2\}$ will be classified. **[25%]**

d) Copy the sketch of the likelihood functions given above and add to it the boundaries between the classification ranges. Indicate on your sketch which regions should be considered to calculate the probability of an error. **[10%]**

Question 1 continued on next page

e) Using the values of $a = 2$ and $b = 3$ and continuing to assume that the priors are equal, calculate the probability of an error. **[25%]**

f) Consider a single feature x with the following likelihoods for two classes:

$$p(x|\omega_1) = \frac{1}{2}, p(x|\omega_2) = \frac{2}{3}\left(1 - \frac{x}{3}\right)$$

The classes are equally likely. The loss, λ_{ij}, is the cost of misclassifying x from class i as belonging to class j. Assume the following loss values for misclassification: $\lambda_{11} = \lambda_{22} = 0$ (no loss for correct classification), $\lambda_{12} = \frac{1}{2}$, and $\lambda_{21} = 1$. Based on the given likelihoods and loss values, what would be the optimal value of the decision boundary x_0 to minimise the risk? **[25%]**

2. a) What are the advantages of the K-Nearest Neighbors (K-NN) classifier compared to parametric classification methods, and what are its possible limitations? **[20%]**

b) Suppose we have a simple 2D dataset with two features: **Height** (in cm) and **Weight** (in kg), and we want to classify whether a person is "Normal BMI" or "High BMI" based on these features. The dataset consists of the following labeled examples:

Height (cm)	Weight (kg)	Class
160	55	Normal BMI
170	70	Normal BMI
175	65	Normal BMI
180	90	High BMI
165	95	High BMI
155	50	Normal BMI

Now, we have a new data point: **Height = 172 cm** and **Weight = 67 kg**, and we want to classify whether this person is "Normal BMI" or "High BMI" using KNN with $K = 3$ and Euclidean distance. Explain the procedure to classify the new data point.

[Hint: The Euclidean distance formula for two points (x_1, y_1) and (x_2, y_2) is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.] **[30%]**

c) We have a dataset of cars with three features: **Engine Size** (L: Litres), **Horsepower** (HP: Horsepower), and **Mileage** (MPG: Miles Per Gallon). The goal is to classify whether a car is **"Luxury"** or **"Economy"**. The dataset is as follow:

Engine Size (L)	Horsepower (HP)	Mileage (MPG)	Class
4.5	450	15	Luxury
2.0	180	30	Economy
3.5	350	20	Luxury
1.8	160	35	Economy
3.0	300	25	Luxury
2.5	200	28	Economy

We want to classify a new car with the following characteristics: **Engine Size = 3.2 L**, **Horsepower = 320 HP**, **Mileage = 22 MPG** using KNN with Manhattan distance and K =4. Explain the procedure to classify the new data point.

[Hint: Manhattan distance: $d = |x_2 - x_1| + |y_2 - y_1| + |z_2 - z_1|$ for two points (x_1, y_1, z_1) and (x_2, y_2, z_2).] **[30%]**

d) The choice of distance measurement in KNN is crucial. What are the main differences between Euclidean distance and Manhattan distance? **[20%]**

3. a) What is the difference of the meaning of K in K-nearest neighbour classifier and K-means clustering? What is the effect of K in K-means clustering? **[20%]**

b) We are given five samples

$$\{(1,3),(2,1),(3,2),(4,-3),(7,4)\}.$$

Please use K-means clustering to cluster the samples into two clusters. The initial cluster centroids are $(1,3)$ and $(2,1)$. The Euclidean distance is the measurement. The complete clustering process is required. **[40%]**

c) If the initial cluster centroids are $(1,3)$ and $(3,2)$ in the previous question, what will be the resulting clustering? **[30%]**

d) What are the limitations of K-means clustering? **[10%]**

4. This question concerns the Perceptron learning algorithm that is used to learn the parameters for classification tasks.

a) Algorithm 1 is an incomplete Perceptron Learning Algorithm pseudo-code. Please complete the missing parts of the pseudo-code.

Algorithm 1 Incomplete Perceptron Learning Algorithm

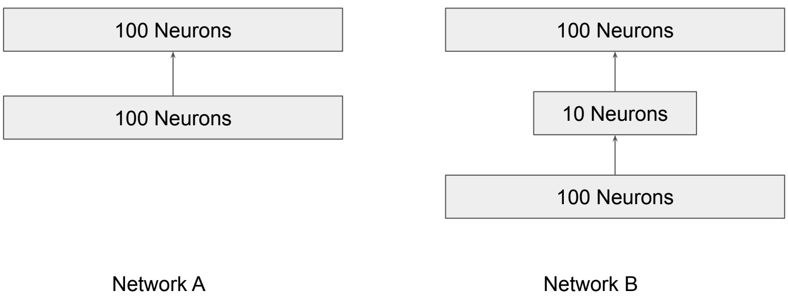
P $\leftarrow$ *inputs with label 1*
N $\leftarrow$ *inputs with label -1*
$\gamma \leftarrow$ *learning rate*
$t = 0$
Initialise W'_t *randomldy*
while Not converge **do**
    Do forward pass
    $\Delta = []$
    for each input $X' \in P \cup N$ **do**
        if $X' \in P$ and $W'^T \cdot X' < 0$ **then**
            !!!Missing Part 1!!!
            !!!Please Complete!!!
        end if
    end for
    !!!Missing Part 2!!!
    !!!Please Complete!!!
    $t+ = 1$
end while

[30%]

b) Consider the following two neural networks, where all of the layers are fully connected.

(i) **Without** considering the bias term, calculate the number of weights for each of the networks. **[10%]**

(ii) Then describe the advantage of Network B over Network A. **[10%]**



Question 4 continued on next page

c) A neural network (as shown in Figure 1) has been designed to model an AND logic gate (as shown in Table 1).

Write down the equation of the neural network in Figure 1 that calculates $\hat{y}$ using a step activation function (f) **[10%]**

x_1	x_2	AND
0	0	0
0	1	0
1	0	0
1	1	1

Table 1: Modelling Boolean Logic for AND Gate

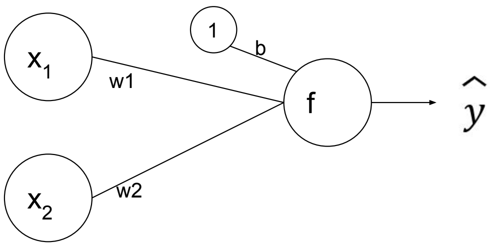


Figure 1: Simple Network

d) The network in **c)** is initialised with the following parameters. $w1 = 0.4$, $w2 = 0.6$ and $b = -0.5$. Is the neural network in Figure 1 able to correctly predict all AND gate outputs (Table 1)? If not, which input will it misclassify? Write down your calculation process. **[20%]**

e) Please apply the perceptron learning algorithm to update the parameters for one iteration using a learning rate $\gamma = 0.4$. After the update, do you think $\gamma = 0.4$ is an appropriate choice for the learning rate? Please justify your answer. **[20%]**

END OF QUESTION PAPER